

MAT 426, Class 15: Householder QR & Orthogonal Diagonalization

When computing matrix-vector and matrix-matrix products involving a Householder matrix H , the matrix H does not have to be explicitly formed since

- $H\mathbf{x} = \left(I - \frac{2\mathbf{u}\mathbf{u}^T}{\mathbf{u}^T\mathbf{u}}\right)\mathbf{x} = \mathbf{x} - \beta\mathbf{u}(\mathbf{u}^T\mathbf{x})$
- $HA^T = A^T - \beta\mathbf{u}(A\mathbf{u})^T = A^T - \beta\mathbf{u}\mathbf{v}^T$ where $\mathbf{v} = A\mathbf{u}$
- $AH = A - \beta A\mathbf{u}\mathbf{u}^T = A - \beta\mathbf{v}\mathbf{u}^T$

THEOREM Given a nonzero vector $\mathbf{x} \neq \mathbf{e}_1$, the Householder matrix H defined by the vector $\mathbf{u} = \mathbf{x} \pm \|\mathbf{x}\|\mathbf{e}_1$ is such that $H\mathbf{x} = \mp\|\mathbf{x}\|\mathbf{e}_1$.

Two Computational Considerations:

- The $\pm$ is chosen by _____ to avoid catastrophic cancellation in computing the first component of $\mathbf{u}$.
- To avoid any possibility of underflow or overflow in the computation of $\|\mathbf{x}\|$ the vector $\mathbf{u}$ can be determined by the vector _____ instead of $\mathbf{x}$ itself. (Our goal is to use this to find the QR decomposition and so we can always rescale our orthonormal vectors in Q without impacting orthogonality.)

Example 1. Let $\mathbf{x} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$. Create a Householder matrix H so that $H\mathbf{x}$ is a multiple of $\mathbf{e}_1$.

QR Factorization via Householder Matrices

Idea: Reduce the matrix A to an _____ matrix R by successively _____ multiplying A with a series of _____ matrices.

Outline of this process for a 4×3 matrix A

Example 2. Find the QR factorization of $A = \begin{bmatrix} 1 & 1 \\ 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}$ using the Householder QR algorithm with four-digit arithmetic.

Full vs Reduced QR Factorization Suppose A is an $m \times n$ matrix.

- The QR factorization via Gram-Schmidt orthogonalization produces an orthogonal matrix Q of size _____ and upper triangular matrix R of size _____. This is called the _____ QR factorization of A .
- The QR factorization via Householder matrices produces an orthogonal matrix Q of size _____ and upper triangular matrix R of size _____. This is called the _____ QR factorization of A .
- How can the full QR factorization $A = QR$ be obtained from the reduced one $A = Q_1 R_1$?

Orthogonal Diagonalization

DIAGONALIZATION THEOREM An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

Example 3. If possible, find a matrix P that diagonalizes $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$.

Example 4. If possible, find a matrix P that diagonalizes $A = \begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}$.

Definition. A square matrix A is **orthogonally diagonalizable** if there exists an _____ matrix Q and a _____ D such that $Q^T A Q = ___$.

THEOREM *If A is orthogonally diagonalizable, then A is symmetric.*

Proof.

THEOREM *If A is a real symmetric matrix, then the eigenvalues of A are real.*

Proof.

THEOREM *If A is a symmetric matrix, then any two eigenvectors corresponding to distinct eigenvalues of A are orthogonal.*

Proof.

THEOREM *Let A be an $n \times n$ real matrix. Then A is symmetric if and only if it is orthogonally diagonalizable.*

Proof.

Example 5. Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.